

NeuroScope

AI-Powered 3D Neural Network Architecture Visualizer and Analyzer

Project Name: NeuroScope — AI-Powered 3D Neural Network Architecture Visualizer and Analyzer

Team Name: DigiNeurons

Category: Education Enhancement

Competition: AYAIR 2026 — Third Edition

Every year, millions of students across Africa begin learning deep learning. They learn to train models, achieve accuracy scores, and deploy applications. But beneath this surface lies a critical gap: most students cannot explain what happens inside the architectures they build. They copy structures from repositories without understanding why a ResNet uses skip connections or why a transformer relies on multi-head attention. When models fail, they cannot diagnose the issue. The network is a black box — they can operate it but never see inside.

This gap is especially harmful in Africa. Egypt's Ministry of Communications has made digital transformation a national priority. The African Union's Agenda 2063 and STISA 2034 strategy call for technically skilled youth to lead the continent's digital industrialization. Yet the tools available to these students were not designed for them. Existing visualizers show only static diagrams, offer no explanation, and provide no feedback on architectural mistakes. A student in Cairo or Nairobi has the same access to model files as one in Silicon Valley — but not the same ability to understand them.

NeuroScope is a web-based application that transforms how students learn deep learning by making neural network architectures visible, interactive, and understandable. A student uploads a model file — supporting ONNX, PyTorch, Keras, and TensorFlow Lite formats — or opens a Python script. NeuroScope parses the code and generates an interactive three-dimensional visualization. Convolutional layers appear as geometric shapes, fully connected layers as flat planes, and attention mechanisms as distinct structures. Clicking any component reveals a plain-language description of its function, the code that created it, and key metrics including parameters, FLOPs, and memory usage.

The architecture health check engine detects eleven common anti-patterns: missing activation functions, sigmoid in deep networks, parameter explosion, redundant layers, missing normalization, and more. Each finding includes severity, explanation, and a suggested fix — functioning as a spell-checker for neural network design. Students receive instant feedback on their architecture without needing a senior engineer to review their work.

NeuroScope integrates into a student's existing workflow through two entry points. The web application runs entirely in the browser with no installation required. It requires no GPU, no cloud credits, and no expensive licenses. It runs on smartphones, tablets, or older laptops — making it accessible to students across Africa's diverse hardware landscape.

The tool supports five model formats: ONNX, PyTorch, Keras, TensorFlow, and TensorFlow Lite. This universality means a student using any major framework can visualize and analyze their model without conversion. The analysis engine computes FLOPs, estimates memory footprint, and predicts training time across different hardware configurations — from a student's laptop to institutional GPU clusters.

NeuroScope directly addresses the mentorship gap in African AI education. When no senior engineer is available to review a student's architecture, NeuroScope provides that review automatically. A first-year student in Port Said can receive the same architectural guidance as a PhD researcher at MIT. This democratization of expertise aligns with the AU's STISA 2034 priority of building scientific literacy, with Agenda 2063's vision of inclusive development, and with Egypt's national digital transformation strategy.

The impact extends beyond individual students. University professors can integrate NeuroScope into curricula, using the 3D visualizations to teach architecture design. Students can contribute new analysis rules and educational descriptions, building a community-driven knowledge base. The tool generates exportable reports — as JSON, PNG diagrams, or text summaries — that students can include in assignments and research papers.

NeuroScope is released under the MIT license as a fully open-source project. Institutions can deploy their own instances. The analysis rules are configurable through YAML files, allowing educators to customize which patterns are detected and at what severity. This extensibility ensures the tool evolves with the field — as new architectures emerge, new rules can be added by the community.

We built NeuroScope not just as a competition entry, but as lasting infrastructure for deep learning education across Africa and beyond. Every learner who struggles to understand why their network is not learning, or what each component in a transformer does, benefits from NeuroScope. By making architecture understanding accessible, we lower the barrier to meaningful participation in AI development.

Word Count: 688